

Ilja Safronovs

R&D Engineer — 3D reconstruction · AI automation · Giraffe360 · Riga, Latvia

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I build 3D Gaussian splatting reconstruction from sparse panoramas at Giraffe360 (PyTorch, CUDA), and AI-agent tools for the team. Four of my results are in production: the windows and mirrors fix, my densification, the exterior orbital videos and the looped drone. For the team I built an agent-driven experiment runner and an MCP server for the task board.

EXPERIENCE

Giraffe360 — R&D Engineer — Riga · 2025 – present

Fixed blurry windows, outdoor views and mirrors in sparse-panorama 3DGS; the production default since 1 June 2026. instagram.com/p/DZsNRB8o772

I ran 140+ controlled training runs, then 600+ experiments over one month to cut haze, floaters and flicker.

Built the sparse-panorama 3DGS training pipeline; my densification runs in production.

The pipeline turns sparse panorama views into 3D Gaussian splats. Jan 2026: influence-accumulation densification, building on Taming 3DGS (Mallick et al., 2024): it measures how much each Gaussian adds to the rendered images and moves the least useful ones; I ported it to the team's new training approach, and it is still used. June 2026: Analytic-Splatting (Liang et al., 2024) in gsplat, with my CUDA forward and backward passes, used in a downsampling step.

Shipped two 3D video features: exterior orbital videos and the looped drone (Oct 2026).

instagram.com/giraffe360/reel/Dd9ODBVIHhE

Exterior: I built the toolkit that orients, selects and crops the outdoor splat, and the per-frame outdoor video pipeline; it makes the orbital view in production. Looped drone: my hackathon idea (Feb 2026, special mention; 3 of my ideas were recognised out of 51); I built its camera-path algorithm.

Mentored a junior engineer on a new depth method; the whole team uses it.

Mostly his work, on denser LiDAR depth.

AI automation: an agent-driven experiment runner, an MCP server for the team task board, a coding-agent harness.

March 2026: an experiment runner, so every teammate can run and track training on a local GPU through an AI agent. June 2026: an MCP server (Node.js), so AI agents can read and write the team task board, with authentication; colleagues use it.

Built an embeddable Rust/wgpu splat viewer and a comparison tool.

A Rust/wgpu viewer (May 2026) that any app can embed and control, and a multi-view comparison tool (Jan 2026).

Prototypes: mirror reflections, meshes from splats, lighter splats.

Working prototypes: correct reflections in mirrors (June 2026), mesh generation from splats, a stage that cuts the splat count for viewers (May 2026), and a fly-in video intro. Feb 2026: I made Difix and NVIDIA's fixer run on the shared GPUs, so the team could evaluate them.

Team tooling: ground-truth validation for splat quality, and one 3DGS monorepo.

July 2026: ground-truth datasets and validation tools, so every teammate can check splat quality against known ground truth. July 2026: I led the restructuring into one 3DGS monorepo, with a code-size check.

Jul–Oct 2025: a small LLM service that wrote tour narration.

It used the OpenAI API and AWS S3.

Giraffe360 makes cameras and software for real-estate virtual tours.

EDUCATION

Riga Technical University — Professional Bachelor, Financial Engineering — 2024 – 2028 (expected)

Probability, statistics, stochastic analysis, Monte Carlo methods, algorithms, databases.

AI Talent Program (Ponder) Nov 2024 – Jan 2025

C++, Python, maths and AI with industry mentors; it led to the Giraffe360 role.

Riga State Classical Gymnasium — Programming and Mathematics track

State mathematics olympiads; Small Mathematics University (University of Latvia).

PROJECTS

Portfolio site: a walkable Gaussian splat of a real flat that I trained (gsplat, 730k splats).

sort_visualization — 16 sorting algorithms in C++/SFML; runs in the browser (WebAssembly).

Offline Graphing Calculator — 4th place of 27 in a school competition.

monte-carlo-finance — option pricing, volatility and risk models in Python.

load-balancing-experiments — seven experiments on DNS, queueing and load balancing.

flow — offline workout tracker in Dart/Flutter, with CI and golden-image tests.

crypto-quant-lab — 59 backtested trading-rule versions on 6 coins.

projects_with_NEAT — neuroevolution agents that learn Flappy Bird and Pong.

RTU_program_parser — finds extra RTU lectures that fit your schedule.

SKILLS

3D & vision	3D Gaussian splatting (gsplat, CUDA kernels), 3D reconstruction from sparse views, computer vision, depth densification, mesh generation, 3D camera paths, splat viewers (Rust/wgpu, three.js/Spark), PyTorch
Research	Reading and implementing papers, controlled experiments, ground-truth evaluation
Languages	Python, C++, Dart/Flutter, JavaScript, Java, SQL
AI automation	LLM APIs (OpenAI), MCP servers (Node.js), agent-driven experiment runners, coding-agent harnesses
Tools	Linux, Docker, Git/GitLab CI, AWS S3, three.js/WebGL, Pygame, Selenium, pandas
Spoken	Latvian, English, Russian